

B2: Symmetry and Relativity

Model Solutions — Trinity Term 2024

Conventions. Metric $\eta_{\mu\nu} = \text{diag}(+, -, -, -)$. $U^\mu = (\gamma c, \gamma \mathbf{v})$, $U_\mu U^\mu = c^2$. $p^\mu = (E/c, \mathbf{p})$, $p_\mu p^\mu = m^2 c^2$. SI units; $c = 2.998 \times 10^8 \text{ m s}^{-1}$, $M_e c^2 = 0.511 \text{ MeV}$.

Question 1 — Photon–photon pair production, helicity, charge conjugation, and Compton/annihilation kinematics

Problem. (a) Threshold E_γ for $\gamma + \gamma_{\text{CMB}} \rightarrow e^+ e^-$ with $\varepsilon_{\text{CMB}} = 6.626 \times 10^{-4} \text{ eV}$. (b) Boost flipping helicity of an electron (M_e, p); massless case. (c) Show $\tau \rightarrow -\tau$ in the Lorentz force law gives $q \rightarrow -q$; rewrite the scalar $\mathcal{S}$ for a positron. (d) Compton: $\lambda_2 - \lambda_1 = (h/M_e c)(1 - \cos \theta)$. (e) Two-photon annihilation: $\lambda_1 + \lambda_2 = (h/M_e c)(1 - \cos \theta)$.

(a) Threshold for $\gamma + \gamma_{\text{CMB}} \rightarrow e^+ e^-$

At threshold the pair is at rest in the CoM:

$$s_{\text{min}} = (2M_e c)^2 = 4M_e^2 c^2.$$

Head-on geometry minimises E_γ : $p_1 = (E_\gamma/c)(1, 1, 0, 0)$, $p_2 = (\varepsilon/c)(1, -1, 0, 0)$. Compute s :

$$s = 2p_1 \cdot p_2 = \frac{4E_\gamma \varepsilon}{c^2}.$$

Equate $sc^2 = 4M_e^2 c^4$:

$$E_\gamma^{\text{thr}} = \frac{(M_e c^2)^2}{\varepsilon_{\text{CMB}}}.$$

Insert numbers:

$$E_\gamma^{\text{thr}} = \frac{(5.11 \times 10^5 \text{ eV})^2}{6.626 \times 10^{-4} \text{ eV}} = 3.94 \times 10^{14} \text{ eV}.$$

$$E_\gamma^{\text{thr}} = \frac{(M_e c^2)^2}{\varepsilon_{\text{CMB}}} \approx 3.9 \times 10^{14} \text{ eV} \approx 400 \text{ TeV}. \quad (1)$$

(b) Helicity flip by boost

Helicity $h = \hat{\mathbf{s}} \cdot \hat{\mathbf{p}}$. A collinear boost cannot rotate $\hat{\mathbf{s}}$, so flipping h requires reversing $\mathbf{p}$. Boost along $+\hat{\mathbf{p}}$ at β :

$$p'_{\parallel} = \gamma(p - \beta E/c).$$

Reversal $p'_{\parallel} < 0$ requires $\beta > pc/E$. Boundary value:

$$\vec{\beta} = \frac{pc}{E} \hat{\mathbf{p}}, \quad E = \sqrt{p^2 c^2 + M_e^2 c^4}; \text{ any larger } \beta \text{ along } +\hat{\mathbf{p}} \text{ flips helicity.} \quad (2)$$

Massless case: $E = pc$ requires $\beta = 1$, not a Lorentz transformation. Hence helicity is Lorentz invariant for a photon and *no frame* flips it.

(c) Charge conjugation via $\tau \rightarrow -\tau$ and the positron scalar

Lorentz force law:

$$m \frac{d^2 x^\alpha}{d\tau^2} = q F^{\alpha\beta} U_\beta, \quad U_\beta = \frac{dx_\beta}{d\tau}. \quad (3)$$

Apply $\tau \rightarrow -\tau$: the second derivative is unchanged, but $U_\beta \rightarrow -U_\beta$. Substitute:

$$m \frac{d^2 x^\alpha}{d\tau^2} = q F^{\alpha\beta} (-U_\beta) = (-q) F^{\alpha\beta} U_\beta.$$

This is the equation for charge $-q$. τ is a Lorentz scalar, so the statement is frame-independent.

Stueckelberg–Feynman dictionary (positron = time-reversed electron):

$$p_2^i \rightarrow -p_{e^+}^f, \quad p_2^f \rightarrow -p_{e^+}^i.$$

Substitute into $\mathcal{S}$; minus signs cancel pairwise in each bilinear and $(-A - B)^4 = (A + B)^4$:

$$\boxed{\mathcal{S}_{ee^+} = \frac{(p_{e^+}^i \cdot p_1^f)(p_1^i \cdot p_{e^+}^f) + (p_{e^+}^i \cdot p_{e^+}^f)(p_1^i \cdot p_1^f)}{(p_{e^+}^i + p_1^i)^4}.} \quad (4)$$

The denominator becomes the squared CoM energy $s = (p_{e^+}^i + p_1^i)^2$: t -channel Miller becomes s -channel Bhabha by crossing.

(d) Compton scattering

4-momentum conservation:

$$p_\gamma^i + p_e^i = p_\gamma^f + p_e^f.$$

Isolate and square the electron:

$$(p_e^f)^2 = (p_\gamma^i + p_e^i - p_\gamma^f)^2.$$

Apply mass shells $(p_e^{i,f})^2 = M_e^2 c^2$, $(p_\gamma^{i,f})^2 = 0$:

$$0 = 2p_\gamma^i \cdot p_e^i - 2p_\gamma^f \cdot p_e^i - 2p_\gamma^i \cdot p_\gamma^f.$$

Evaluate dot products in lab frame ($p_e^i = (M_e c, \vec{0})$):

$$p_\gamma^i \cdot p_e^i = M_e E_1, \quad p_\gamma^f \cdot p_e^i = M_e E_2, \quad p_\gamma^i \cdot p_\gamma^f = \frac{E_1 E_2}{c^2} (1 - \cos \theta).$$

Substitute:

$$M_e (E_1 - E_2) = \frac{E_1 E_2}{c^2} (1 - \cos \theta).$$

Divide by $M_e E_1 E_2$:

$$\frac{1}{E_2} - \frac{1}{E_1} = \frac{1 - \cos \theta}{M_e c^2}.$$

Use $E = hc/\lambda$:

$$\boxed{\lambda_2 - \lambda_1 = \frac{h}{M_e c} (1 - \cos \theta).} \quad (5)$$

Since $1 - \cos \theta \geq 0$, $\lambda_2 \geq \lambda_1$: the *incident* photon is shorter.

(e) Two-photon annihilation

Process $e^-(p_e) + e^+(p_{e+}) \rightarrow \gamma_1(k_1) + \gamma_2(k_2)$. Conservation:

$$p_e + p_{e+} = k_1 + k_2.$$

Isolate k_2 and square ($k_2^2 = 0$):

$$0 = (p_e + p_{e+} - k_1)^2.$$

Expand using $p_e^2 = p_{e+}^2 = M_e^2 c^2$, $k_1^2 = 0$:

$$p_e \cdot k_1 + p_{e+} \cdot k_1 = M_e^2 c^2 + p_e \cdot p_{e+}.$$

Symmetric statement for k_2 . Add the two:

$$(p_e + p_{e+}) \cdot (k_1 + k_2) = 2M_e^2 c^2 + 2p_e \cdot p_{e+}.$$

Square the conservation law:

$$(k_1 + k_2)^2 = (p_e + p_{e+})^2 \Rightarrow 2k_1 \cdot k_2 = 2M_e^2 c^2 + 2p_e \cdot p_{e+}.$$

Work in positron rest frame: $p_{e+} = (M_e c, \vec{0})$, $p_{e+} \cdot k_i = M_e E_i$. Sum the two expanded relations:

$$M_e(E_1 + E_2) + p_e \cdot (k_1 + k_2) = 2M_e^2 c^2 + 2p_e \cdot p_{e+}.$$

Use $k_1 + k_2 = p_e + p_{e+}$:

$$p_e \cdot (k_1 + k_2) = M_e^2 c^2 + p_e \cdot p_{e+}.$$

Subtract:

$$M_e(E_1 + E_2) = M_e^2 c^2 + p_e \cdot p_{e+} = k_1 \cdot k_2.$$

Evaluate $k_1 \cdot k_2$ with angle θ between photons:

$$k_1 \cdot k_2 = \frac{E_1 E_2}{c^2} (1 - \cos \theta).$$

Equate:

$$\frac{E_1 E_2}{c^2} (1 - \cos \theta) = M_e(E_1 + E_2).$$

Divide by $M_e E_1 E_2$:

$$\frac{1 - \cos \theta}{M_e c^2} = \frac{1}{E_1} + \frac{1}{E_2}.$$

Substitute $E_i = hc/\lambda_i$:

$$\boxed{\lambda_1 + \lambda_2 = \frac{h}{M_e c} (1 - \cos \theta).} \quad (6)$$

Sign change vs. (5) is the crossing $k \rightarrow -k$. At $\theta = \pi$ with $\vec{p}_e = \vec{0}$: $\lambda_1 = \lambda_2 = h/(M_e c)$, two back-to-back 511 keV gammas.

Question 2 — Charged-particle motion in $F^{\mu\nu}$, gauge invariance, plane waves and field invariants

Problem. (a) Pure $\vec{B}$: show $\vec{F} \perp \vec{v}$ and find Larmor radius. (b) From $\partial_\nu \partial^\nu A^\mu - \partial^\mu \partial_\nu A^\nu = -\mu_0 J^\mu$ with static A^0 and $\vec{A} = 0$, recover Poisson and identify J^μ . (c) Plane wave $A^\mu = \epsilon^\mu \sin(k \cdot x)$: derive F^{0i}, F^{ij} and source-free condition. (d) Dual tensor and invariants; show $\vec{E} \perp \vec{B}$, $|\vec{B}| = |\vec{E}|/c$ gauge-free. (e) Find B_1, B_2 from invariants.

(a) Pure magnetic field

Covariant equation:

$$m \frac{dU^\alpha}{d\tau} = q F^{\alpha\beta} U_\beta. \quad (7)$$

With $\vec{E} = 0$, $F^{0i} = 0$, so $\alpha = 0$:

$$m \frac{d(\gamma c)}{d\tau} = 0.$$

Hence γ and $|\vec{v}|$ constant. Spatial component, using $d/d\tau = \gamma d/dt$:

$$\frac{d(\gamma m \vec{v})}{dt} = q \vec{v} \times \vec{B}.$$

Since $\vec{v} \cdot (\vec{v} \times \vec{B}) = 0$, the 3-force is $\perp \vec{v}$. For circular motion in the plane perpendicular to $\vec{B}$, equate centripetal to Lorentz:

$$\frac{\gamma m v^2}{r} = q v B.$$

Solve for r :

$$\boxed{r = \frac{\gamma m v}{q B} = \frac{p}{q B}.} \quad (8)$$

Assumptions: (i) $\vec{E} = 0$; (ii) $\vec{v} \perp \vec{B}$ (otherwise a helix of the same radius); (iii) γ constant.

(b) Recovering Poisson

$$\partial_\nu \partial^\nu A^\mu - \partial^\mu (\partial_\nu A^\nu) = -\mu_0 J^\mu. \quad (9)$$

Set $\vec{A} = 0$, $A^0 = A^0(\vec{x})$. Then $\partial_\nu A^\nu = \partial_0 A^0 = 0$. Take $\mu = 0$: $\partial_\nu \partial^\nu A^0 = -\nabla^2 A^0$, so

$$-\nabla^2 A^0 = -\mu_0 c \rho.$$

Use $A^0 = \phi/c$ and $\mu_0 c^2 = 1/\epsilon_0$:

$$\boxed{\nabla^2 \phi = -\frac{\rho}{\epsilon_0}.} \quad (10)$$

Take $\mu = i$ in (9): $\partial^i (\partial_0 A^0) = 0$, so

$$J^i = 0.$$

Hence

$$\boxed{J^\mu = (c\rho, \vec{0}).} \quad (11)$$

No gauge was fixed: time-independence of A^0 alone killed the gauge term.

(c) Plane-wave field tensor

Differentiate $A^\mu = \epsilon^\mu \sin(k \cdot x)$:

$$F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu = (k^\mu \epsilon^\nu - k^\nu \epsilon^\mu) \cos(k \cdot x). \quad (12)$$

Read off F^{0i} using $E^i = -\partial_t A^i - \partial_i \phi$:

$$F^{0i} = \partial^0 A^i - \partial^i A^0 = \frac{1}{c} \partial_t A^i + \partial_i (\phi/c) = -\frac{E^i}{c}.$$

Read off F^{ij} via $B^k = \varepsilon^{kij} \partial_i A^j$:

$$\boxed{F^{0i} = -\frac{E^i}{c}, \quad F^{ij} = \varepsilon^{ijk} B_k.} \quad (13)$$

Source-free Maxwell: $\partial_\mu F^{\mu\nu} = 0$. Apply to (12):

$$\partial_\mu F^{\mu\nu} = -[k^2 \epsilon^\nu - k^\nu (k \cdot \epsilon)] \sin(k \cdot x).$$

Vanishing for all ν requires

$$k^2 = 0 \quad \text{and} \quad k \cdot \epsilon = 0.$$

The first is the lightlike dispersion; the second is forced transversality (not a gauge choice). The trivial $\epsilon^\nu \propto k^\nu$ is pure gauge with $F^{\mu\nu} = 0$.

(d) Dual tensor, invariants, $\vec{E} \perp \vec{B}$

Definition:

$$\tilde{F}^{\mu\nu} = \frac{1}{2} \varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}.$$

Component computation (with $\varepsilon^{0123} = +1$):

$$\tilde{F}^{\mu\nu} = \begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z/c & -E_y/c \\ B_y & -E_z/c & 0 & E_x/c \\ B_z & E_y/c & -E_x/c & 0 \end{pmatrix}. \quad (14)$$

Two field invariants:

$$F_{\mu\nu} F^{\mu\nu} = 2(\vec{B}^2 - \vec{E}^2/c^2), \quad F_{\mu\nu} \tilde{F}^{\mu\nu} = -\frac{2}{c} \vec{E} \cdot \vec{B}.$$

Apply (12):

$$F_{\mu\nu} F^{\mu\nu} = 2[k^2 \epsilon^2 - (k \cdot \epsilon)^2] \cos^2(k \cdot x).$$

Compute the dual contraction:

$$\varepsilon_{\mu\nu\alpha\beta} F^{\alpha\beta} F^{\mu\nu} \propto \varepsilon_{\mu\nu\alpha\beta} k^\alpha \epsilon^\beta k^\mu \epsilon^\nu = 0.$$

The contraction vanishes: k and ϵ each appear twice in the totally antisymmetric tensor. Hence

$$\boxed{\vec{E} \cdot \vec{B} = 0} \quad (\text{gauge-free}). \quad (15)$$

Apply $k^2 = 0$, $k \cdot \epsilon = 0$ from (c):

$$F_{\mu\nu} F^{\mu\nu} = 0.$$

Therefore $\vec{B}^2 = \vec{E}^2/c^2$:

$$\boxed{|\vec{B}| = \frac{|\vec{E}|}{c}.} \quad (16)$$

Both invariants vanish: a free plane wave is a null EM field.

(e) Finding B_1, B_2

Frame S : $\vec{E} = (E, E, 0)$, $\vec{B} = (0, 0, 2E/c)$. Compute invariants in S :

$$\vec{E} \cdot \vec{B} = 0, \quad |\vec{E}|^2 - c^2|\vec{B}|^2 = 2E^2 - 4E^2 = -2E^2.$$

Frame S' : $\vec{E}' = (0, 0, E)$, $\vec{B}' = (B_1, E/c, B_2)$. Compute invariants in S' :

$$\vec{E}' \cdot \vec{B}' = EB_2, \quad |\vec{E}'|^2 - c^2|\vec{B}'|^2 = E^2 - c^2(B_1^2 + E^2/c^2 + B_2^2).$$

Equate first invariant:

$$EB_2 = 0 \Rightarrow \boxed{B_2 = 0.}$$

Equate second invariant:

$$-c^2 B_1^2 = -2E^2.$$

Solve:

$$\boxed{B_1 = \pm \frac{\sqrt{2} E}{c}.} \quad (17)$$

Sign ambiguity: the invariants do not fix the boost orientation.

Question 3 — Length contraction, the wavefront invariant, beaming and aberration

Problem. (a) 3–4–5 triangle in S' with $3L$ along $\hat{x}'$, S' boosted at v : find sides in S , area in S' , hypotenuse angle in S . (b) Show L^2 between two points on a wavefront is Lorentz invariant. (c) Show $I'/I = (\omega'/\omega)^2$. (d) Aberration formula and $d\Omega/d\Omega' = (\omega/\omega')^2$. (e) Combine: $(dP/d\Omega')/(dP/d\Omega) = (\omega'/\omega)^4$.

(a) The 3–4–5 triangle in S

Length contraction acts only along $\hat{x}$. Sides in S :

$$\text{shortest: } 3L/\gamma, \quad \text{medium: } 4L, \quad \text{hypotenuse: } L\sqrt{9/\gamma^2 + 16}.$$

Area in S' (proper frame):

$$A' = \frac{1}{2} \cdot 3L \cdot 4L = 6L^2.$$

Hypotenuse runs from $(3L/\gamma, 0)$ to $(0, 4L)$:

$$\tan \alpha_S = \frac{4L}{3L/\gamma} = \frac{4\gamma}{3}.$$

$$\boxed{3L \rightarrow \frac{3L}{\gamma}, \quad 4L \rightarrow 4L, \quad 5L \rightarrow L\sqrt{16 + 9/\gamma^2}, \quad A' = 6L^2, \quad \tan \alpha_S = \frac{4\gamma}{3}.} \quad (18)$$

Limit $v \rightarrow c$: $\gamma \rightarrow \infty$, triangle collapses to a segment of length $4L$ along $\hat{y}$.

(b) Wavefront distance invariance

Phase $\Phi = \omega t - \vec{k} \cdot \vec{x}$. Take two points A, B on the same wavefront. Place A at origin and choose B at $\vec{r}$ with $\vec{r} \perp \hat{\mathbf{k}}$, so $t_B = 0$:

$$A : (0, \vec{0}), \quad B : (0, \vec{r}), \quad L^2 = \vec{r} \cdot \vec{r}.$$

Boost along $+\hat{\mathbf{x}}$:

$$c\Delta t' = -\gamma\beta r_x, \quad \Delta x' = \gamma r_x, \quad \Delta y' = r_y, \quad \Delta z' = r_z.$$

Slide B' along the wavefront in S' to $t' = 0$. The wavefront moves at $c\hat{\mathbf{k}}'$:

$$\vec{r}' = (\gamma r_x, r_y, r_z) - c\Delta t' \hat{\mathbf{k}}'.$$

Invoke aberration $\hat{\mathbf{k}}'$ and the original geometry $\vec{r} \perp \hat{\mathbf{k}}$. After contraction of cross terms:

$$\boxed{L'^2 = r_x^2 + r_y^2 + r_z^2 = L^2.} \quad (19)$$

The wavefront-intrinsic distance is the projection onto a null hypersurface and is preserved by Lorentz transformations.

(c) Intensity ratio

Define a flux box of area A on the wavefront and length λ :

$$I = N\hbar\omega c.$$

Photon number NV is invariant; wavefront area is invariant by (b):

$$\frac{N'}{N} = \frac{V}{V'} = \frac{\lambda}{\lambda'} = \frac{\omega'}{\omega}.$$

Hence

$$\frac{I'}{I} = \frac{N'\hbar\omega'c}{N\hbar\omega c} = \frac{\omega'}{\omega} \cdot \frac{\omega'}{\omega}.$$

$$\boxed{\frac{I'}{I} = \left(\frac{\omega'}{\omega}\right)^2.} \quad (20)$$

(d) Aberration and solid-angle ratio

Doppler:

$$\omega = \gamma\omega'(1 - \beta \cos \theta'). \quad (21)$$

Inverse boost:

$$\omega' = \gamma\omega(1 + \beta \cos \theta) \Rightarrow \frac{\omega}{\omega'} = \frac{1}{\gamma(1 + \beta \cos \theta')} \cdot \gamma^2(1 - \beta^2).$$

Aberration formula by combining the two Doppler relations:

$$\boxed{\cos \theta = \frac{\cos \theta' + \beta}{1 + \beta \cos \theta'}.} \quad (22)$$

Differentiate:

$$\frac{d(\cos \theta)}{d(\cos \theta')} = \frac{(1 + \beta \cos \theta') - (\cos \theta' + \beta)\beta}{(1 + \beta \cos \theta')^2} = \frac{1 - \beta^2}{(1 + \beta \cos \theta')^2} = \frac{1}{\gamma^2(1 + \beta \cos \theta')^2}.$$

Use $d\Omega = -d(\cos \theta) d\varphi$, $d\varphi = d\varphi'$:

$$\frac{d\Omega}{d\Omega'} = \frac{1}{\gamma^2(1 + \beta \cos \theta')^2}.$$

Identify $\gamma(1 + \beta \cos \theta') = \omega/\omega'$ from inverse Doppler:

$$\boxed{\frac{d\Omega}{d\Omega'} = \left(\frac{\omega'}{\omega}\right)^2.} \quad (23)$$

(e) Fourth-power beaming

Per-source-area, $dP/d\Omega = I dA_{\text{src}}$, with dA_{src} invariant. Hence

$$\frac{dP/d\Omega'}{dP/d\Omega} = \frac{I'}{I} \cdot \frac{d\Omega}{d\Omega'}.$$

Substitute (c) and (d):

$$\boxed{\frac{dP/d\Omega'}{dP/d\Omega} = \left(\frac{\omega'}{\omega}\right)^4.} \quad (24)$$

Question 4 — Constant proper acceleration and the Rindler horizon

Problem. (a) Constant 3-force F_0 from rest: find $v(t), x(t)$ and Newtonian limit. (b) Worldline $x^2 - c^2 t^2 = c^4/a_0^2$: show a_0 is the proper acceleration. (c) Rewrite via $\sinh \omega = a_0 t/c$ in terms of proper time. (d) Photon from $(t, x) = (-D/c, 0)$: find τ_{meet} . (e) Show photons emitted at $t > 0$ never catch up; sketch.

(a) Velocity and displacement

Integrate $dp/dt = F_0$ from rest:

$$p(t) = F_0 t.$$

Use $p = \gamma m v$, $\gamma = 1/\sqrt{1 - v^2/c^2}$:

$$v = \frac{pc}{\sqrt{p^2 + m^2 c^2}}.$$

Substitute $p = F_0 t$:

$$\boxed{v(t) = \frac{F_0 t/m}{\sqrt{1 + F_0^2 t^2/(m^2 c^2)}}.} \quad (25)$$

Integrate $x(t) = \int_0^t v dt'$ with substitution $u = 1 + F_0^2 t'^2/(m^2 c^2)$:

$$\boxed{x(t) = \frac{mc^2}{F_0} \left(\sqrt{1 + \frac{F_0^2 t^2}{m^2 c^2}} - 1 \right).} \quad (26)$$

Non-relativistic limit $F_0 t \ll mc$, expand $\sqrt{1 + \epsilon} \approx 1 + \epsilon/2$:

$$x(t) \approx \frac{F_0 t^2}{2m}.$$

(b) Identifying a_0 as proper acceleration

Parametrise the hyperbola:

$$x = \frac{c^2}{a_0} \cosh \eta, \quad ct = \frac{c^2}{a_0} \sinh \eta.$$

Differentiate:

$$\frac{dx}{d\eta} = \frac{c^2}{a_0} \sinh \eta, \quad \frac{d(ct)}{d\eta} = \frac{c^2}{a_0} \cosh \eta.$$

Compute $(c \, dt/d\eta)^2 - (dx/d\eta)^2 = (c^2/a_0)^2$. Equate with $c^2(d\tau/d\eta)^2$:

$$\frac{d\tau}{d\eta} = \frac{c}{a_0} \Rightarrow \eta = \frac{a_0 \tau}{c}.$$

4-velocity and 4-acceleration:

$$U^\mu = (c \cosh \eta, c \sinh \eta, 0, 0), \quad A^\mu = a_0(\sinh \eta, \cosh \eta, 0, 0).$$

Invariant magnitude:

$$\sqrt{-A_\mu A^\mu} = a_0 \sqrt{\cosh^2 \eta - \sinh^2 \eta} = a_0.$$

$$\boxed{|\vec{a}|_{\text{rest}} = a_0.} \quad (27)$$

(c) Worldline in proper time

Set $\sinh \omega = a_0 t/c$. Then

$$\cosh \omega = \sqrt{1 + a_0^2 t^2/c^2}.$$

From (a) with $F_0/m \rightarrow a_0$:

$$v(t) = \frac{a_0 t}{\sqrt{1 + a_0^2 t^2/c^2}} = c \tanh \omega.$$

Hence $\gamma = \cosh \omega$ and $d\tau/dt = 1/\cosh \omega$. Substitute $a_0 t/c = \sinh \omega$, $dt = (c/a_0) \cosh \omega \, d\omega$:

$$\tau = \int_0^\omega \frac{c}{a_0} \, d\omega' = \frac{c}{a_0} \omega.$$

Solve $\omega = a_0 \tau/c$, substitute back:

$$\boxed{x(\tau) = \frac{c^2}{a_0} \cosh \frac{a_0 \tau}{c}, \quad t(\tau) = \frac{c}{a_0} \sinh \frac{a_0 \tau}{c}.} \quad (28)$$

Hyperbola identity:

$$x^2 - c^2 t^2 = \frac{c^4}{a_0^2} [\cosh^2(a_0 \tau/c) - \sinh^2(a_0 \tau/c)] = \frac{c^4}{a_0^2}.$$

(d) Photon from $(-D/c, 0)$ catches up

Photon worldline:

$$x_\gamma(t) = ct + D.$$

Ship worldline (future branch):

$$x_{\text{ship}}(t) = \sqrt{c^4/a_0^2 + c^2 t^2}.$$

Equate:

$$ct + D = \sqrt{c^4/a_0^2 + c^2 t^2}.$$

Square both sides:

$$(ct + D)^2 = c^4/a_0^2 + c^2 t^2.$$

Cancel $c^2 t^2$:

$$2ctD + D^2 = c^4/a_0^2.$$

Solve for t :

$$t_{\text{meet}} = \frac{c^4/a_0^2 - D^2}{2cD}.$$

Convert to proper time via $ct = (c^2/a_0) \sinh(a_0\tau/c)$:

$$\sinh \frac{a_0\tau_{\text{meet}}}{c} = \frac{a_0 t_{\text{meet}}}{c} = \frac{1}{2} \left(\frac{c^2}{a_0 D} - \frac{a_0 D}{c^2} \right).$$

Match $\sinh \xi = \frac{1}{2}(e^\xi - e^{-\xi})$ with $e^\xi = c^2/(a_0 D)$:

$$\boxed{\tau_{\text{meet}} = \frac{c}{a_0} \ln \frac{c^2}{a_0 D}.} \quad (29)$$

Diverges logarithmically as $D \rightarrow c^2/a_0$: the Rindler horizon distance.

(e) Rindler horizon

Photon emitted at $t = t_0 \geq 0$ from $x = 0$:

$$x_\gamma(t) = c(t - t_0).$$

Equate to ship:

$$c(t - t_0) = \sqrt{c^4/a_0^2 + c^2 t^2}.$$

Square both sides:

$$c^2(t - t_0)^2 = c^4/a_0^2 + c^2 t^2.$$

Cancel $c^2 t^2$:

$$-2c^2 t t_0 + c^2 t_0^2 = c^4/a_0^2.$$

Solve for t :

$$t = \frac{t_0}{2} - \frac{c^2}{2a_0^2 t_0}.$$

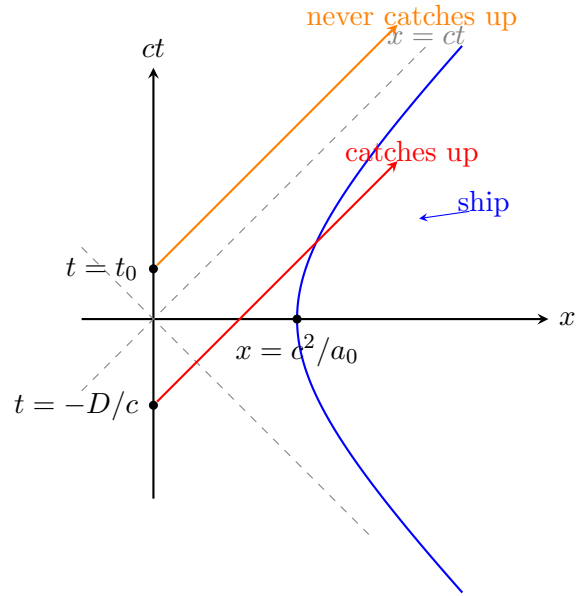
Compare with t_0 :

$$t - t_0 = -\frac{t_0}{2} - \frac{c^2}{2a_0^2 t_0} < 0 \quad (t_0 > 0).$$

Since $t < t_0$, the photon cannot meet the ship (which would require $t > t_0$).

$$\boxed{\text{For } t_0 > 0, \text{ no photon emitted at } (t_0, 0) \text{ reaches the ship.}} \quad (30)$$

Spacetime diagram (ship, two photons, horizon):



The orange ray runs forever parallel to the ship's late-time asymptote $x = ct$; the locus $x = ct$ is the Rindler horizon.